

3.1.2 Monotone Sequences

Definition: A sequence $\{a_n\}_{n=1}^{\infty}$

- is called increasing if $a_n \leq a_{n+1}$ for all $n \geq 1$. It is called strictly increasing if $a_n < a_{n+1}$ for all $n \geq 1$.
- is called decreasing if $a_n \geq a_{n+1}$ for all $n \geq 1$. It is called strictly decreasing if $a_n > a_{n+1}$ for all $n \geq 1$.
- A sequence that is increasing or decreasing is called monotone (or monotonic)
- A sequence that is strictly increasing or strictly decreasing is called strictly monotone.

Example 1: Prove that the sequence $a_n = \frac{n}{n+1}$ is strictly increasing.

A calculus flavored proof is as follows.

Proof: Consider the function $f(x) = \frac{x}{x+1}$. We note that $f'(x) = \frac{(x+1) - x}{(x+1)^2} = \frac{1}{(x+1)^2}$.

Now, if $x \geq 1$, we have that $f'(x) > 0$. Thus, we know from calculus 1 that $f(x) = \frac{x}{x+1}$ is

increasing on $(1, \infty)$. So, for any natural number n with $n \geq 1$, we have

$f(n) < f(n+1) \Leftrightarrow a_n < a_{n+1}$. Thus, the sequence is increasing by definition. ■

It is also possible to prove this without calculus. Consider the following proof of the same result.

Proof: We note the difference between consecutive terms:

$$a_{n+1} - a_n = \frac{n+1}{(n+1)+1} - \frac{n}{n+1} = \frac{(n+1)(n+1)}{(n+2)(n+1)} - \frac{n(n+2)}{(n+1)(n+2)} = \frac{n^2 + 2n + 1 - n^2 - 2n}{(n+1)(n+2)} = \frac{1}{(n+1)(n+2)}$$

Now, we can see that if $n \geq 1$ then $a_{n+1} - a_n > 0$. This implies that $a_{n+1} > a_n$ for all n , so we have that the sequence is increasing by definition. ■

It is also possible to prove this using a ratio of consecutive terms.

Proof: We note that:

$$\frac{a_{n+1}}{a_n} = \frac{\left(\frac{n+1}{(n+1)+1} \right)}{\left(\frac{n}{n+1} \right)} = \frac{n+1}{n+2} \cdot \frac{n+1}{n} = \frac{n^2 + 2n + 1}{n^2 + 2n}$$

Now, we can see that if $n \geq 1$ then $\frac{a_{n+1}}{a_n} > 1$. This implies that $a_{n+1} > a_n$ for all n , so we have that the sequence is increasing by definition. ■

Note: Each of the proof techniques shown in this example can be useful in the right context.

Definition: If discarding finitely many terms from the beginning of a sequence produces a sequence with a certain property, then the original sequence is said to have that property eventually.

Example 2:

Show that the sequence $a_n = \left\{ \frac{10^n}{n!} \right\}_{n=1}^{\infty}$ is eventually strictly decreasing.

For small values of n we can see that the sequence is actually increasing (observe $n=1$ & $n=2$).

However we know that eventually $n!$ produces much larger values than 10^n so the sequence should start to decrease at some point.

We will look at a ratio of consecutive terms to determine when this happens.

$$\frac{a_{n+1}}{a_n} = \frac{\left(\frac{10^{n+1}}{(n+1)!} \right)}{\left(\frac{10^n}{n!} \right)} = \frac{10^{n+1}}{(n+1)!} \cdot \frac{n!}{10^n} = \frac{10}{n+1}$$

We can see that for $n \geq 10$, $\frac{a_{n+1}}{a_n} < 1$ which implies that $a_{n+1} < a_n$. Therefore we see that the sequence a_n is strictly decreasing for $n \geq 10$ and therefore by definition a_n is eventually strictly decreasing. ■

Definition: A sequence $\{a_n\}$ is bounded above if there is a number M such that $a_n \leq M$ for all $n \geq 1$.

- M is called an upper bound for the sequence $\{a_n\}$

A sequence $\{a_n\}$ is bounded below if there is a number m such that $a_n \geq m$ for all $n \geq 1$.

- m is called a lower bound for the sequence $\{a_n\}$

A sequence $\{a_n\}$ is a bounded sequence if it has an upper & a lower bound.

Example 3: Show that the sequence $a_n = 6 + \frac{1}{n}$ is a bounded sequence

a_n is never larger than 7 since $\frac{1}{n}$ is never larger than 1 for $n \geq 1$. Therefore an upper bound for a_n is 7.

a_n is never smaller than 6 since $\frac{1}{n}$ is always larger than 0 for $n \geq 1$. Therefore a lower bound for a_n is 6.

Since a_n has an upper and a lower bound it is a bounded sequence. ■

Theorem (Monotonic Sequence Theorem):

Every bounded above, increasing (or eventually increasing) sequence is convergent.

Every bounded below, decreasing (or eventually decreasing) sequence is convergent.

More often stated as:

Every bounded, monotone (or eventually monotone) sequence is convergent.

The proof of this theorem requires some additional content knowledge about the Completeness Axiom.

We will not discuss the proof in class, however details can be found on pages 611 & 612 of the textbook.

Note: This theorem is very valuable in establishing whether a sequence is convergent. It says nothing about what a sequence converges to.

Example 4 (you try): Consider the sequence $a_n = \frac{1}{n}$

(a) Prove that this sequence is decreasing.

We note that for positive n , $\frac{a_{n+1}}{a_n} = \frac{\left(\frac{1}{n+1}\right)}{\left(\frac{1}{n}\right)} = \frac{n}{n+1} < 1$. Therefore we have $a_n > a_{n+1}$ for $n \geq 1$, and the sequence

is decreasing by definition.

(b) Show that this sequence is bounded below.

This sequence is bounded below by 0 since $a_n = \frac{1}{n} > 0$ for all $n \geq 1$.

(c) What can you conclude from parts (a) and (b) by the Monotonic Sequence Theorem?

We may conclude that a_n is convergent (this means $\lim_{n \rightarrow \infty} a_n$ exists as a finite number).

Note: One might think that the end result from the above example is quite weak, and I would agree. The main purpose of this example was to show how the monotonic sequence theorem could be applied in a simple situation. For a nontrivial application of this theorem consider the following problem:

Consider the sequence $\left\{ \sqrt{2}, \sqrt{2\sqrt{2}}, \sqrt{2\sqrt{2\sqrt{2}}}, \dots \right\}$. Determine if the sequence converges.

Example 5: Show that the sequence $a_n = \left\{ \frac{10^n}{n!} \right\}_{n=1}^{\infty}$ converges and find its limit.

In example 2 we showed that a_n was eventually decreasing, so it is eventually monotone and since all the terms of the sequence are positive the sequence has a lower bound of $m = 0$. Therefore by the monotone sequence theorem a_n converges to some value L .

Now that we know that a_n converges we must find what it converges to. We will have to be a little clever.

Note that:

$$a_{n+1} = \frac{10^{n+1}}{(n+1)!} = \frac{10}{n+1} \cdot \frac{10^n}{n!} = \frac{10}{n+1} a_n$$

We will take the limit as $n \rightarrow \infty$ of both sides and use the fact that $\lim_{n \rightarrow \infty} a_{n+1} = \lim_{n \rightarrow \infty} a_n = L$.

We then obtain:

$$\begin{aligned} \lim_{n \rightarrow \infty} a_{n+1} &= \lim_{n \rightarrow \infty} \frac{10}{n+1} a_n \\ \Leftrightarrow L &= \lim_{n \rightarrow \infty} \frac{10}{n+1} \cdot \lim_{n \rightarrow \infty} a_n \\ \Leftrightarrow L &= 0 \cdot L \\ \Leftrightarrow L &= 0 \end{aligned}$$

Therefore we have found that $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{10^n}{n!} = L = 0$.

■

In general we can adapt the strategy used above to obtain the useful fact that $\lim_{n \rightarrow \infty} \frac{x^n}{n!} = 0$. This fact will very useful in chapter 4.